



Architecture Competence

The ideal architect should be a man of letters, a skillful draftsman, a mathematician, familiar with historical studies, a diligent student of philosophy, acquainted with music, not ignorant of medicine, learned in the responses of jurisconsults, familiar with astronomy and astronomical calculations.

—Vitruvius, *De Architectura* (25 B.C.)

The lyf so short, the craft so long to lerne.

—Geoffrey Chaucer

If software architecture is worth “doing,” surely it’s worth doing well. Most of the literature about architecture concentrates on the technical aspects. This is not surprising; it is a deeply technical discipline. There is little information that speaks to the fact that architectures are created by architects working in organizations, full of actual human beings. Dealing with these humans is decidedly non-technical. What can be done to help architects, especially architects-in-training, be better at this important dimension of their job? And what can be done to help architecture organizations do a better job in fostering their architects to produce their best work?

An organization’s ability to routinely produce high-quality architectures that are aligned with its business goals well cannot be understood simply through examination of past architectures and measurement of their deficiencies. The organizational and human causes of those deficiencies also need to be understood.

This chapter is about the competence of individual architects and the organizations that wish to produce high-quality architects. We define the architecture competence of an organization as follows:

The architecture competence of an organization is the ability of that organization to grow, use, and sustain the skills and knowledge

necessary to effectively carry out architecture-centric practices at the individual, team, and organizational levels to produce architectures with acceptable cost that lead to systems aligned with the organization's business goals.

Because the architecture competence of an organization depends, in part, on the competence of architects, we begin by asking what it is that architects are expected to do, know, and be skilled at. Then we'll look at what organizations can and should do to help their architects produce better architectures. Individual and organizational competencies are intertwined. Studying only one or the other won't do.

24.1 Competence of Individuals: Duties, Skills, and Knowledge of Architects

Architects perform many activities beyond directly producing an architecture. These activities, which we call *duties*, form the backbone of an individual's architecture competence. We surveyed a broad body of information aimed at architects (such as websites, courses, books, and position descriptions for architects). We also surveyed practicing architects. These surveys tell us that duties are but one aspect of individual competence. Writers about architects also speak of *skills* and *knowledge*. For example, the ability to communicate ideas clearly and to negotiate effectively are skills often ascribed to competent architects. In addition, architects need to have up-to-date knowledge about patterns, database platforms, web services standards, quality attributes, and a host of other topics.

Duties, skills, and knowledge¹ form a triad upon which architecture competence for individuals rests. The relationship among these three is shown in Figure 24.1—namely, skills and knowledge support the ability to perform the required duties. Infinitely talented architects are of no use if they cannot (for whatever reason) perform the duties required of the position; we would not say they were competent.

To give examples of these concepts:

- “Design the architecture” is a duty.
- “Ability to think abstractly” is a skill.
- “Patterns and tactics” is a part of the body of knowledge.

This example purposely illustrates that skills and knowledge are important (only) for supporting the ability to carry out duties effectively. As another example, “documenting the architecture” is a duty, “ability to write clearly” is a skill, and “ISO Standard 42010” is part of the related body of knowledge. Of course, a skill or knowledge area can support more than one duty.

1. Some writers speak of the importance of experience. We count experience as a form of knowledge.

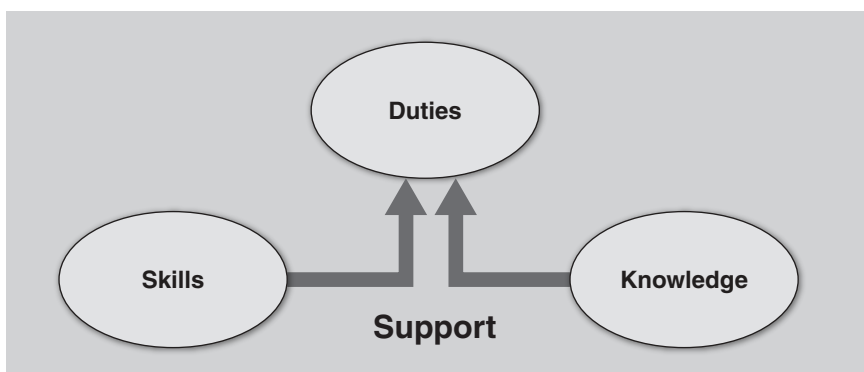


FIGURE 24.1 Skills and knowledge support the execution of duties.

Knowing the duties, skills, and knowledge of architects (or, more precisely, the duties, skills, and knowledge that are needed of architects in a particular organizational setting) can help establish measurement and improvement strategies for individual architects. If you want to improve your individual architectural competence, you should do the following:

1. *Gain experience carrying out the duties.* Apprenticeship is a productive path to achieving experience. Education alone is not enough, because education without on-the-job application merely enhances knowledge.
2. *Improve your nontechnical skills.* This dimension of improvement involves taking professional development courses, for example, in leadership or time management. Some people will never become truly great leaders or communicators, but we can all improve on these skills.
3. *Master the body of knowledge.* One of the most important things a competent architect must do is master the body of knowledge and remain up to date on it. To emphasize the importance of remaining up to date, consider the advances in knowledge required for architects that have emerged in just the last few years. For example, the cloud and edge computing that we discuss in Chapters 26 and 27 were not important topics several years ago. Taking courses, becoming certified, reading books and journals, visiting websites and portals, reading blogs, attending architecture-oriented conferences, joining professional societies, and meeting with other architects are all useful ways to improve knowledge.

Duties

This section summarizes a wide variety of an architect's duties. Not every architect in every organization will perform every one of these duties on every project. But a competent architect should not be surprised to find himself or herself engaged in

any of the activities listed here. We divide the duties into technical duties (Table 24.1) and nontechnical duties (Table 24.2). One immediate observation you should make is how many nontechnical duties there are. An obvious implication, for those of you who wish to be architects, is that you must pay adequate attention to the nontechnical aspects of your education and your professional activities.

TABLE 24.1 The Technical Duties of a Software Architect

General Duty Area	Specific Duty Area	Example Duties
Architecting	Creating an architecture	Design or select an architecture. Create software architecture design plan. Build product line or product architecture. Make design decisions. Expand detail and refine design to converge on final design. Identify the patterns and tactics and articulate the principles and key mechanisms of the architecture. Partition the system. Define how the components fit together and interact.
	Evaluating and analyzing an architecture	Evaluate an architecture (for your current system or for other systems) to determine satisfaction of use cases and quality attribute scenarios. Create prototypes. Participate in design reviews. Review construction-level designs. Review the designs of the components designed by junior engineers. Review designs for compliance with the architecture. Compare software architecture evaluation techniques. Apply value-based architecting techniques to evaluate architectural decisions. Model alternatives. Perform tradeoff analysis.
	Documenting an architecture	Prepare architectural documents and presentations useful to stakeholders. Document software interfaces. Produce documentation standards. Document variability and dynamic behavior.
	Working with and transforming existing system(s)	Maintain and evolve existing system and its architecture. Redesign existing architecture(s) for migration to new technology and platforms.
	Performing other architecting duties	Sell the vision, keep the vision alive. Participate in product design meetings. Give technical advice on architecture, design, and development. Provide architectural guidelines for software design activities. Lead architecture improvement activities. Participate in software process definition and improvement. Define philosophy and principles for global architecture. Oversee or manage the architecture definition process. Provide architecture oversight of software development projects.

General Duty Area	Specific Duty Area	Example Duties
Duties concerned with life-cycle activities other than architecting	Managing the requirements	Analyze functional and quality attribute software requirements. Understand business and customer needs and ensure that the requirements meet these needs. Capture customer, organizational, and business requirements on the architecture. Create software specifications from business requirements. Articulate and refine architectural requirements. Listen to and understand the scope of the project. Understand the client's key design needs and expectations.
	Implementing the product	Produce code. Conduct code reviews. Develop reusable software components. Analyze, select, and integrate software components. Set and ensure adherence to coding guidelines. Recommend development methodologies and coding standards. Monitor, mentor, and review the work of outside consultants and vendors.
	Testing the product	Establish architecture-based testing procedures. Support system testers. Support field testing. Support bug fixing and maintenance.
	Evaluating future technologies	Evaluate and recommend enterprise's software solutions. Manage the introduction of new software solutions. Analyze current IT environment and recommend solutions for deficiencies. Work with vendors to represent organization's requirements and influence future products. Develop and present technical white papers.
	Selecting tools and technology	Perform technical feasibility studies of new technologies and architectures. Evaluate commercial tools and software components from an architectural perspective. Develop internal technical standards and contribute to the development of external technical standards.

Skills

Given the wide range of duties enumerated in the previous section, what skills (beyond mastery of the technical body of knowledge) does an architect need to possess? Much has been written about the architect's special role of leadership in a project; the ideal architect is an effective communicator, manager, team builder, visionary, and mentor. Some certificate or certification programs emphasize non-technical skills. Common to these certification programs are nontechnical assessment areas of leadership, organization dynamics, and communication.

Table 24.3 enumerates the set of nontechnical skills most useful to an architect.

TABLE 24.2 The Nontechnical Duties of a Software Architect

General Duty Area	Specific Duty Area	Example Duties
Management	Managing the project	Help with budgeting and planning. Follow budgetary constraints. Manage resources. Perform sizing and estimation. Perform migration planning and risk assessment. Take care of or oversee configuration control. Create development schedules. Measure results using quantitative metrics and improve both personal results and teams' productivity. Identify and schedule architectural releases.
	Managing the people	Build "trusted advisor" relationships. Coordinate. Motivate. Advocate. Delegate. Act as a supervisor.
	Supporting the management	Provide feedback on appropriateness and difficulty of project. Advise the project manager on the tradeoffs between software design choices and requirements choices. Provide input to software project manager in the software project planning and estimation process. Serve as a "bridge" between the technical team and the project manager.
Organization and business-related duties	Supporting the organization	Grow an architecture evaluation capability in the organization. Review and contribute to research and development efforts. Participate in the hiring process for the team. Help with product marketing. Institute and oversee cost-effective software architecture design reviews. Help develop intellectual property.
	Supporting the business	Translate business strategy into technical vision and strategy. Influence the business strategy. Understand and evaluate business processes. Understand and communicate the business value of software architecture. Help the organization meet its business goals. Understand customer and market trends. Identify, understand, and resolve business issues. Align architecture with the business goals and objectives.
Leadership and team building	Providing technical leadership	Mentor other architects. Produce technology trend analysis or roadmaps.
	Building a team	Set team context (vision). Build the architecture team and align them with the vision. Mentor junior architects. Educate the team on the use of the architecture. Maintain morale, both within and outside the architecture group. Foster the professional development of team members. Coach teams of software design engineers for planning, tracking, and completion of work within the agreed plan. Mentor and coach staff in the use of software technologies. Work both as a leader and as an individual contributor.

TABLE 24.3 The Nontechnical Skills of a Software Architect

General Skill Area	Specific Skill Area	Example Skills
Communication skills	Outward	Ability to make oral and written communications and presentations. Ability to present and explain technical information to diverse audiences. Ability to transfer knowledge. Ability to persuade. Ability to see from and sell to multiple viewpoints.
	Inward	Ability to listen, interview, consult, and negotiate. Ability to understand and express complex topics.
Interpersonal skills	Within team	Ability to be a team player. Ability to work effectively with superiors, colleagues, and customers. Ability to maintain constructive working relationships. Ability to work in a diverse team environment. Ability to inspire creative collaboration. Ability to build consensus.
	With other people	Ability to demonstrate interpersonal skills. Ability to be diplomatic and respect others. Ability to mentor others. Ability to handle and resolve conflict.
Work skills	Leadership	Ability to make decisions. Ability to take initiative and be innovative. Ability to demonstrate independent judgment, be influential, and command respect.
	Workload management	Ability to work well under pressure, plan, manage time, estimate. Ability to support a wide range of issues and work on multiple complex projects concurrently. Ability to effectively prioritize and execute tasks in a high-pressure environment.
	Skills to excel in corporate environment	Ability to think strategically. Ability to work under general supervision and under given constraints. Ability to organize workflow. Ability to sense where the power is and how it flows in an organization. Ability to do what it takes to get the job done. Ability to be entrepreneurial, be assertive without being aggressive, and receive constructive criticism.
	Skills for handling information	Ability to be detail oriented while maintaining overall vision and focus. Ability to see the big picture. Ability to deal with abstraction.
	Skills for handling the unexpected	Ability to tolerate ambiguity. Ability to take and manage risks. Ability to solve problems. Ability to be adaptable, flexible, open minded, and resilient. Ability to do the juggling necessary to deploy successful software projects.

Knowledge

A competent architect has an intimate familiarity with the architectural body of knowledge. The software architect should

- Be comfortable with all branches of software engineering from requirements definition to implementation, development, verification and validation, and deployment
- Be familiar with supporting disciplines such as configuration management and project management
- Understand current design and implementation tools and technologies

Knowledge and experience in one or more application domains is also necessary.

Table 24.4 is the set of knowledge areas for an architect.

TABLE 24.4 The Knowledge Areas of a Software Architect

General Knowledge Area	Specific Knowledge Area	Specific Knowledge Examples
Computer science knowledge	Knowledge of architecture concepts	Knowledge of architecture frameworks, architectural patterns, tactics, viewpoints, standard architectures, relationship to system and enterprise architecture, architecture description languages, emerging technologies, architecture evaluation models and methods, and quality attributes.
	Knowledge of software engineering	Knowledge of systems engineering. Knowledge of software development life cycle, software process management, and improvement techniques. Knowledge of requirements analysis, mathematics, development methods and modeling techniques, elicitation techniques. Knowledge of component-based software development, reuse methods and techniques, software product-line techniques, documentation, testing and debugging tools.
	Design knowledge	Knowledge of different tools and design techniques. Knowledge of how to design complex multi-product systems. Knowledge of object-oriented analysis and design, UML diagrams, and UML analysis modeling.
	Programming knowledge	Knowledge of programming languages and programming language models. Knowledge of specialized programming techniques for security, real time, etc.

General Knowledge Area	Specific Knowledge Area	Specific Knowledge Examples
Knowledge of technologies and platforms	Knowledge of specific technologies and platforms	Knowledge of hardware/software interfaces, web-based applications, and Internet technologies. Knowledge of specific software/operating systems, such as RDBMS concepts, cloud platforms, and SOA implementations.
	General knowledge of technologies and platforms	Knowledge of IT industry future directions and the ways in which infrastructure impacts an application.
Knowledge about the organization's context and management	Domain knowledge	Knowledge of the most relevant domain(s) and domain-specific technologies.
	Industry knowledge	Knowledge of the industry's best practices and industry standards. Knowledge of how to work in onshore/offshore team environment.
	Enterprise knowledge	Knowledge of the company's business practices, and your competition's products, strategies, and processes. Knowledge of business and technical strategy, and business reengineering principles and processes. Knowledge of strategic planning, financial models, and budgeting.
	Leadership and management techniques	Knowledge of coaching, mentoring, and training software developers. Knowledge of project management. Knowledge of project engineering.

24.2 Competence of a Software Architecture Organization

Organizations by their practices and structure can help or hinder architects in performing their duties. For example, if an organization has a career path for architects, that will motivate employees to become architects. If an organization has a standing architecture review board, then the project architect will know how and with whom to schedule a review. Lack of these items will mean that an architect has to fight battles with the organization or determine how to carry out a review without internal guidance. It makes sense, therefore, to ask whether a particular organization is architecturally competent and to develop instruments whose goal is measuring the architectural competence of an organization. The architectural competence of organizations is the topic of this section.

Activities Carried Out by a Competent Organization

Organizations have duties, skills, and knowledge for architecture as well. For example, adequately funding the architecture effort is an organizational duty, as is effectively using the available architecture workforce (by appropriate teaming, and so on). These are organizational duties because they are outside the control of individual architects. An organization-level skill might be effective knowledge management or human resource management as applied to architects. An example of organizational knowledge is the composition of an architecture-based life-cycle model that software projects may employ.

Here are some things—duties—that an organization can perform to help improve the success of its architecture efforts:

- Hire talented architects.
- Establish a career track for architects.
- Make the position of architect highly regarded through visibility, reward, and prestige.
- Establish a clear statement of responsibilities and authority for architects.
- Establish a mentoring program for architects.
- Establish an architecture training and education program.
- Establish an architect certification program.
- Have architects receive external architect certifications.
- Measure architects' performance.
- Establish a forum for architects to communicate and share information and experience.
- Establish a repository of reusable architectures and architecture-based artifacts.
- Develop reusable reference architectures.
- Establish organization-wide architecture practices.
- Establish an architecture review board.
- Measure the quality of architectures produced.
- Provide a centralized resource to analyze and help with architecture tools.
- Hold an organization-wide architecture conference.
- Have architects join professional organizations.
- Bring in outside expert consultants on architecture.
- Include architecture milestones in project plans.
- Have architects provide input into product definition.
- Have architects advise on the development team structure.
- Give architects influence throughout the entire project life cycle.
- Reward or penalize architects based on project success or failure.

Assessment Goals

The activities enumerated above can be assessed. What are the potential goals from an assessment of an organization's architecture competence? There are at least four sets of reasons for assessing organizational architectural competence:

1. There are goals relevant to any business that wishes to improve their architectural competence. Businesses regularly assess their own performance in a variety of means—technical, fiscal, operational (for example, consider the widespread use of multi-criteria techniques such as the Balanced Scorecard or Business/IT Alignment in industry)—for a variety of reasons. These include determining whether they are meeting industry norms and gauging their progress over time in meeting organizational goals.
2. There are goals relevant to an acquisition organization. For example, an organization can use an assessment of architecture competence to assess a contractor in much the same way that contractors are scrutinized with respect to their CMMI level. Or an organization might use an assessment of architecture competence to aid in deciding among competing bids from contractors. All other things being equal, an acquiring organization would prefer a contractor with a higher level of architectural competence because this typically means fewer downstream problems and rework. An acquisition organization might assess the contractors directly, or hire a third party to do the assessment.
3. There are goals relevant to service organizations: such organizations might be motivated to maintain, measure, and advertise their architectural competence as a means of attracting and retaining customers. In such a case they would typically rely on outside organizations to assess their level of competence in an objective fashion.
4. Finally, there are goals that are relevant to product builders: these organizations would be motivated to assess, monitor (and, over time, increase) their level of architectural competence as it would (1) aid in advertising the quality of their products and (2) aid in their internal productivity and predictability. In fact, in these ways their motivations are aligned with those of service organizations.

Assessing an Organization's Competence

In addition to duties, skills, and knowledge, there are other models of individual and organizational competence that are helpful in building an instrument for assessing an organization's competence. They are the following:

- The *Human Performance Technology model*, which measures the value of an individual or department's output and the cost of producing that output.

It holds that competent people produce the most value for every organizational dollar spent.

- The *Organizational Coordination model*, which measures how teams in multiple sites developing a single product or related set of products cooperate to produce a functioning product. An organization that is architecturally competent will have more effective and efficient coordination mechanisms than an organization that is not architecturally competent.
- The *Organizational Learning model*, which measures how well an organization's learning processes transform experience into knowledge, moderated by context.

We have created a framework for organizational architecture competence that forms the foundation for a competence assessment procedure. A small team of trained assessors can use the framework to conduct interviews (with architects, developers, and technical and organizational managers), examine current practices, read documents and evidentiary artifacts (such as organizational standards), and investigate architecture-based successes and failures in the recent past. They can use their findings to identify systemic trouble spots and recommend improvement strategies.

Table 24.5 shows the framework. For convenience, it is divided into practice areas that relate to *software and system engineering*, *technical management* (which is by and large the management of single projects or small numbers of related projects), and *organizational management* (which is management at a scope more broad than that of projects).

The framework is populated by questions inspired by the four models of competence that we previously described. Each question has a set of answers that we might expect to see in a competent organization. For example, a question associated with the practice area “Hire talented architects” deals with how a candidate architect's experience and capabilities are assessed. Expected answers might include having the architect take a test, requiring that candidates possess an architecture certification, or examining previous architectures designed by the candidate. (Our expected answers grow as we visit more and more organizations. It's a pleasant surprise when we find an organization carrying out a practice area in a clever way that we hadn't thought of.)

Questions Based on the Duties, Skills, and Knowledge Model. Our assessment framework contains dozens of questions related to duties, skills, and knowledge. The questions are posed in terms of the organization: The questions ask how the organization ensures that the architectural duties are carried out in a competent manner, and how the organization measures and nurtures its architects' skills and knowledge. Here is a small set of example questions based on the Duties, Skills, and Knowledge model (chosen from among the dozens that populate our assessment framework) that we use in an architecture competence assessment exercise with an organization.

TABLE 24.5 Framework for Organizational Architecture Competence

Software Engineering Practice Areas	Quality Attribute Elicitation Practices
	Tools and Technology Selection
	Modeling and Prototyping Practices
	Architecture Design Practices
	Architecture Description Practices
	Architecture Evaluation Practices
	System Implementation Practices ▪ Software design practices (design conforms to architecture) ▪ Software coding practices (code conforms to design and architecture)
	Software Verification Practices ▪ Proving properties of the software ▪ Software testing
	Architecture Reconstruction Practices
Technical Management Practice Areas	Business or Mission Goals Practices ▪ Setting goals ▪ Measuring achievement of organization's goals ▪ Performance-based compensation
	Product or System Definition Practices ▪ Setting functional requirements
	Allocating Resources ▪ Setting architect's workload and schedule ▪ Funding stakeholder involvement
	Project Management Practices ▪ Project plan structure aligned with architecture structure ▪ Adequate time planned for architecture evaluation
	Process Discipline Practices ▪ Establish organization-wide architecture practices ▪ Process monitoring and improvement practices ▪ Reuse practices
	Collaboration with Manager Practices ▪ Architects advise managers ▪ Architects support managers

continues

TABLE 24.5 Framework for Organizational Architecture Competence, *continued*

Organizational Management Practice Areas	Hire Talented Architects
	Establish a Career Track for Architects
	▪ Leadership roles for architects
	▪ Succession planning
	Professional Development Practices
	▪ Ongoing training
	▪ Creating and sustaining an internal community of architects
	▪ Supporting participation in external communities
	Organizational Planning Practices
	Technology Planning and Forecasting Practices

Duty: Creating an Architecture

Question: How do you create an architecture?

- How do you ensure that the architecture is aligned with the business goals?
- What is the input into the architecture creation process? What inputs are provided to the architect?
- How does the architect validate the information provided? What does the architect do in case the input is insufficient or inadequate?

Duty: Architecture Evaluation and Analysis

Question: How do you evaluate and analyze an architecture?

- Are evaluations part of the normal software development life cycle or are they done when problems are encountered?
- Is the evaluation incremental or “big bang”? How is the timing determined?
- Does the evaluation include an explicit activity relating architecture to business goals?
- What are the inputs to the evaluation? How are they validated?
- What are the outputs from an evaluation? How are the outputs of the evaluation utilized? Are the outputs differentiated according to impact or importance? How are the outputs validated? Who is communicated what outputs?

Knowledge: Architecture Concepts

Question: How does your organization ensure that its architects have adequate architectural knowledge?

- How are architects trained in general knowledge of architecture?
- How do architects learn about architectural frameworks, patterns, tactics, standards, documentation notations, and architecture description languages?

- How do architects learn about new or emerging architectural technologies (e.g., multi-core processors)?
- How do architects learn about analysis and evaluation techniques and methods?
- How do architects learn quality attribute-specific knowledge, such as techniques for analyzing and managing availability, performance, modifiability, and security?
- How are architects tested to ensure that their level of knowledge is adequate, and remains adequate, for the tasks that they face?

Questions Based on the Organizational Coordination Model. Questions based on the Organizational Coordination model focus on how the organization establishes its teams and what support it provides for those teams to coordinate effectively. Here are a couple of example questions:

Question: How is the architecture designed with distribution of work to teams in mind?

- How available or broadly shared is the architecture to various teams?
- How do you manage the evolution of architecture during development?
- Is the work assigned to the teams before or after the architecture is defined, and with due consideration of the architectural structure?

Question: Are the aspects of the architecture that will require a lot of inter-team coordination supported by the organization's coordination/communication infrastructure?

- Do you co-locate teams with high coordination? Or at least put them in the same time zone?
- Must all coordination among teams go through the architecture team?

Questions Based on the Human Performance Technology Model.

The Human Performance Technology questions deal with the value and cost of the organization's architectural activities. Here are examples of questions based on the Human Performance Technology model:

Question: Do you track how much the architecture effort costs, and how it impacts overall project cost and schedule?

- How do you track the end of architecture activities?
- How do you track the impact of architecture activities?

Question: Do you track the value or benefits of the architecture?

- How do you measure stakeholder satisfaction?
- How do you measure quality?

Questions Based on the Organizational Learning Model. Finally, a set of example questions, based on the Organizational Learning model, which deal with how the organization systematically internalizes knowledge to its advantage:

Question: How do you capture and share experiences, lessons learned, technological decisions, techniques and methods, and knowledge about available tooling?

- Do you use any knowledge management tools?
- Is capture and use of architectural knowledge embedded in your processes?
- Where is the information about “who knows what” captured and how is this information maintained?
- How complete and up to date is your architecture documentation? How widely disseminated is it?

Performing an Assessment

Our organizational competence assessment is carried out using a team of three to four assessors. The exercise is set up by establishing the scope of the review: Are we assessing the entire company? One of its divisions? Or perhaps a single important project?

After we establish the scope, we identify the groups we wish to interview. Of course, we'll want to interview the architecture team(s) within the scope. From there we identify groups both upstream and downstream of the architects. Upstream are groups that manage the architects or provide organization-wide architecture training. Downstream, we interview the “consumers” of the architectures, such as developers, integrators, testers, and maintainers. We interview small groups, making sure that no members of an interview group have reporting relationships with each other.

We try hard to establish an informal atmosphere in the group interviews, to avoid inhibiting the participants. The tone is conversational, not inquisitorial. We begin each interview by reminding the participants of the purpose of the exercise, and to assure them that nothing they say will be quoted to anyone outside the group in any way that could identify them.

For each group, we have planned which parts of the framework we wish to discuss with that group. We won't ask testers, for example, questions intended for managers, and vice versa. We use the questions as a guide for the conversation, but not a rigid script. Whenever we pose a question in the assessment instrument, there are a number of meta-questions that automatically accompany it. For example:

- What evidence could you show us to support your answer? Supporting evidence might include a software development plan that lays out the role of architecture in a project, an organization's architecture-based training curriculum, or many other kinds of documentation.
- How sustainable is your answer over time, over different systems, and across different architects? For example, we might ask how an answer might change if a different architect came on board the project.

The outcome of an assessment is organized by the practice areas of the framework. For each practice, we assign one of three values that correspond to “you’re doing this well,” “you could be doing this better (and experiencing more benefit),” and “this is an area of high risk.” Graphically, we show this as green light, yellow light, red light. We have found that this simple metric provides organizations with enough granularity to turn their attention to problem areas, which is the whole point of the assessment. We do not give an overall rating. Thus, we closely mirror the “continuous representation” option of maturity models such as CMMI, in which the result is a vector rather than a scalar.

We present the findings in a written report and a slide presentation. In both, we describe and justify each finding, based on what we were told in the interviews and/or read in provided documents.

24.3 Summary

The vast majority of work on software and systems architecture (including our own) has focused on the technical aspects. But an architecture is much more than a technical “blueprint” for a system. This has led us to try to understand, in a more holistic way, what an architect and an architecture-centric organization must do to succeed. To this end, we have developed a framework that aids us in assessing an organization for competence.

We use the framework to ask questions about an organization’s practices. We can also ask about recent architecture successes and failures, and investigate the causes of each. The output of this exercise is a formal report that assesses competence at organization, team, and individual levels. Along with this report we make improvement recommendations based on assessment results; these, too, are tied to the underlying competence models.

You can do the same sort of evaluation on your own organization. The key to the process is in understanding the various models and in creating questions based on these models that aid you in assessing how well you are doing in those areas that you care about. Given this knowledge, you can create your own improvement plan, as an individual architect or for an entire organization.

24.4 For Further Reading

The four models that underlie the assessment framework presented here are described in more detail in the Technical Note “Models for Evaluating and Improving Architecture Competence” [Bass 08]. These models are the following:

- *Duties, Skills, and Knowledge (DSK) model of competence.* This model is predicated on the belief that architects and architecture-producing organizations are useful sources for understanding the tasks necessary to the job of architecting. To assemble a comprehensive set of duties, skills, and knowledge for architects, we surveyed approximately 200 sources of information targeted to professional architects—books, websites, blogs, position descriptions, and more. The results of this survey can be found in [Clements 07].
- *Human Performance model of competence.* This model is based on the human performance engineering work of Thomas Gilbert [Gilbert 07]. This model is predicated on the belief that competent individuals in any profession are the ones who produce the most valuable results at a reasonable cost. Using this model will involve figuring out how to measure the value and cost of the outputs of architecture efforts, finding areas where that ratio can be improved, and crafting improvement strategies based on environmental and behavioral factors.
- *Organizational Coordination model of competence.* The focus of this model is on creating an interteam coordination model for teams developing a single product or a closely related set of products. The architecture for the product induces a requirement for teams to coordinate during the realization or refinement of architectural decisions. The organizational structure, practices, and tool environment of the teams allow for particular types of coordination with a particular interteam communication bandwidth. The coordination model of competence compares the requirements for coordination that the architecture induces with the bandwidth for coordination supported by the organizational structure, practices, and tool environment [Cataldo 07].
- *Organizational Learning model of competence.* This model is based on the concept that organizations, and not just individuals, can learn. Organizational learning is a change in the organization that occurs as a function of experience. This change can occur in the organization's cognitions or knowledge (e.g., as presented by Fiol and Lyles [Fiol 85]), its routines or practices (e.g., as demonstrated by Levitt and March [Levitt 88]), or its performance (e.g., as presented by Dutton and Thomas [Dutton 84]). Although individuals are the medium through which organizational learning generally occurs, learning by individuals within the organization does not necessarily imply that organizational learning has occurred. For learning to be organizational, it has to have a supra-individual component [Levitt 88]. There are three approaches to measure organizational learning: (1) measure knowledge directly through questionnaires, interviews, and verbal protocols; (2) treat changes in routines and practices as indicators of changes in knowledge; or (3) view changes in organizational performance indicators associated with experience as reflecting changes in knowledge [Argote 07].

The Open Group offers a certification program for qualifying the skills, knowledge, and experience of IT, business, and enterprise architects, which is related to measuring and certifying an individual architect's competence. Visit opengroup.org for details. The International Association of Software Architects (IASA) offers a similar certification; see iasahome.org.

Dana Bredemeyer and Ruth Malan have written many articles on the role of the software architect (www.bredemeyer.com/who.htm), including their duties and skills [Bredemeyer 11] (www.bredemeyer.com/Architect/ArchitectSkillsLinks.htm). They have their own competence framework as well as a skills development program.

The U.K. Chapter of the International Council on Systems Engineering (INCOSE) maintains a “Core Competencies Framework” for systems engineers that includes a “Basic Skills and Behaviours” section listing “the usual common attributes required by any professional engineer” [INCOSE 05]. The list includes coaching, communication, negotiation and influencing, and “team working.”

The classic work on the Balanced Scorecard was created by Kaplan and Norton [Kaplan 92] and the classic work on Business/IT Alignment was originally created by Luftman [Luftman 00], although this has been updated to explicitly consider the role of architecture in alignment [Chen 10].

Boehm, Valerdi, and Honour [Boehm 07] provide one of the few empirical studies of systems engineering and how an investment in “better” engineering pays off (or doesn't) in the future.

24.5 Discussion Questions

1. In which skills and knowledge discussed in this chapter do you think you might be deficient? How would you reduce these deficiencies?
2. Which duties, skills, or knowledge do you think are the most important or cost-effective to improve in an individual architect? Justify your answer.
3. How would you measure the specific value of architecture in a project? How would you distinguish the value added by architecture from the value added by other activities such as quality assurance or configuration management?
4. How do you measure items such as “customer satisfaction” or “negotiation skills”? How would you validate such measurements?
5. How would you distinguish benefits caused by systematic organizational learning from the benefits due to heroic efforts by individuals within the organization?

6. Section 24.2 listed a number of practices of an architecturally competent organization. Prioritize that list based on expected benefit over expected cost.
7. Suppose you are in charge of hiring an architect for an important system in your company. How would you go about it? What would you ask the candidates in an interview? Would you ask them to produce anything? If so, what? Would you have them take a test of some kind? If so, what? Who in your company would you have interview them? Why?